b) **Digging :** Clay dug out from ground is spread on level ground about 60 cm to 120 cm heaps.

c) **Cleaning :** Stones, pebbles, vegetable matter, etc., are removed and the lumps are converted in to powder form.

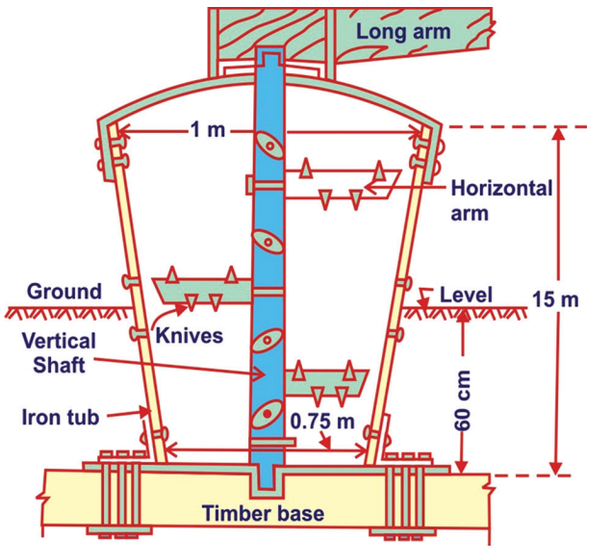
d) **Weathering :** Clay is to be exposed to atmosphere from few weeks to full season.

e) **Blending :** Clay is made loose and any ingredient to be added to it is spread out at top and it is turned up and down in vertical direction.

f) **Tempering:** Clay is brought to a proper degree of hardness. Then water is added to the clay and the whole mass is kneaded or pressed under the feet of men or cattle.



For large scale, tempering is usually done in pug mill as shown in the fig.



Pug Mill

Process : Clay with water is placed in pug mill from the top. When the vertical shaft is rotated by using electric power, clay is thoroughly mixed up by the actions of horizontal arms and knives. When clay has been sufficiently pugged, hole at the bottom of tub is opened and the pugged earth is taken out from ramp.

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3.2.5.2. Moulding

Clay which is prepared from pug mill is sent to the next operation of moulding.

Following are the two ways of moulding :

1. Hand Moulding
2. Machine Moulding.

1) Hand Moulding :

Moulds are rectangular boxes made of wood or steel which are open at top and bottom. Steel moulds are more durable and used for manufacturing bricks on large scale as shown in fig. Bricks prepared by hand moulding are of two types.

a) Ground Moulding
b) Table Moulding

a) Ground Moulding :

Ground is first made level and fine sand is sprinkled over it. Mould is dipped in water and placed over the ground to fill the clay. Extra clay is removed by wooden or metal strike, after the mould is filled. Mould is then lifted up and raw brick is left on the ground. Mould is then dipped in water every time. Lower faces of

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